
EE1060: Differential Equations and Transform Techniques

Lecture 10: Equilibrium points and Stability of ODE

Topics :

1. First Order Autonomous Systems
 2. Equilibrium points and Stability
-

Trajectory, Autonomous Differential Equations, Examples

independent Variable $\rightarrow$ time (t)
 dependent Variable $\rightarrow$ 'y'

Solving ODE involving $\frac{dy}{dt}$

$$\frac{dy}{dt} = f(y) \quad \left\{ \begin{array}{l} \rightarrow \text{can be linear} \\ \text{Non-linear} \end{array} \right. \quad \leftarrow \text{Autonomous DE}$$

Example:-

$$Ri + L \frac{di}{dt} = V_{dc} \Rightarrow \frac{di}{dt} = \underbrace{\frac{1}{L} (V_{dc} - Ri)}_{f(i)}$$

for an Autonomous DE $y=c$ is an isocline. with level value $f(c)$.

is it a solution?

Analysis of Solutions of Autonomous ODE - Equilibrium/Critical Points

Set of points where $f(y) = 0$ (y^*) constitute the Equilibrium points

↳ steady state.

the trajectory of $\frac{dy}{dt} = f(y)$ with $y_0 = y^*$ is a straight line.
 ← separable DE | solve for analytical solution by integrating on both sides.

Example:-

$$\frac{di}{dt} = \frac{1}{L} (V_{dc} - Ri) \rightarrow \text{Equilibrium points} = \frac{V_{dc}}{R}$$

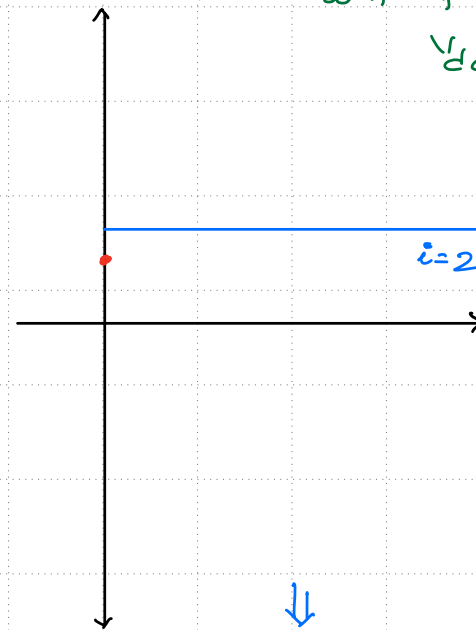
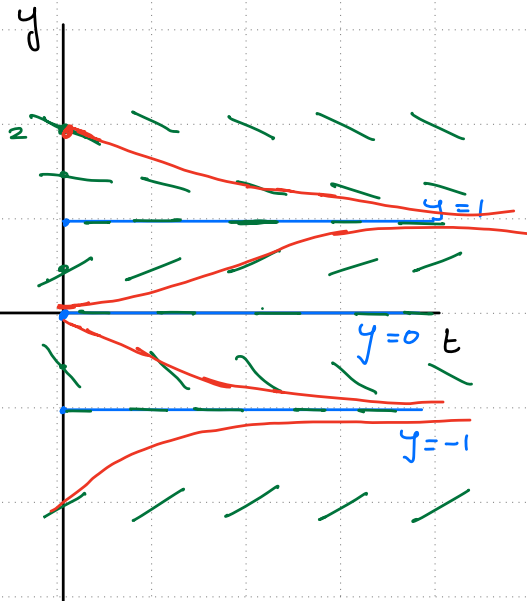
$$\frac{dy}{dt} = y - y^3 \rightarrow \text{Equilibrium points: } (0, 1, -1).$$

$$= \mu y - y^3 \rightarrow \text{Equilibrium points: } (0, \sqrt{\mu}, -\sqrt{\mu})$$

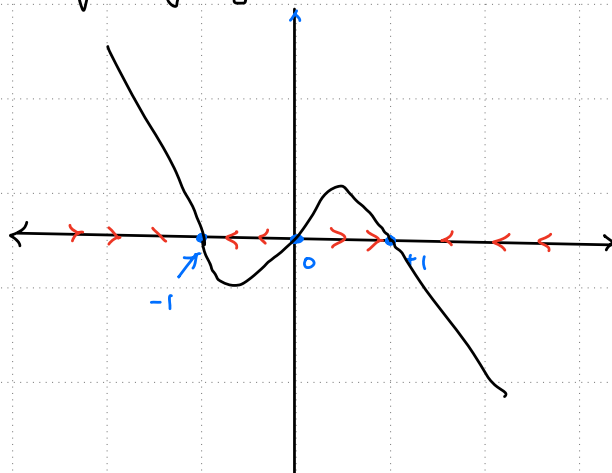
Analysis of Solutions of Autonomous ODE - Two Ways of Denoting the Solutions

Analyze the solution of $\frac{dy}{dt} = f(y)$ with init cond. y_0 .

Rh. Circ. Loop $R, L = 1$
 $V_{dc} = 2$

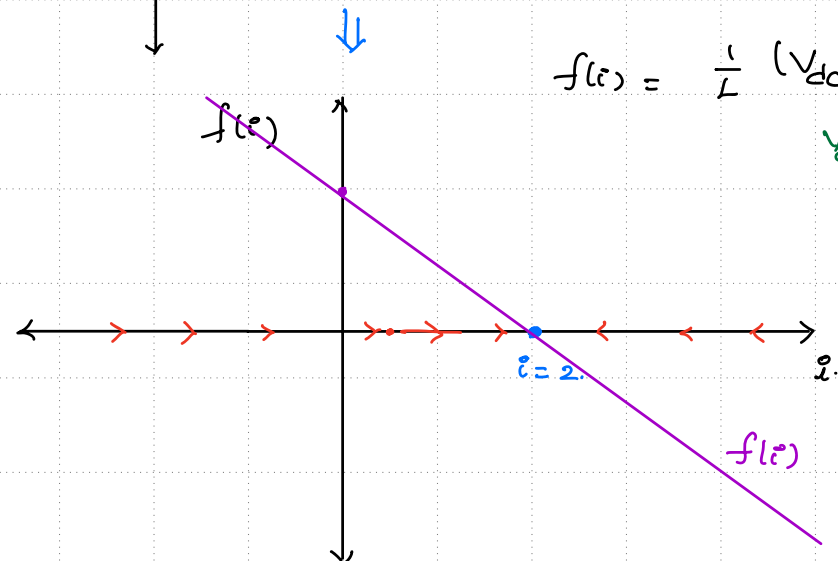


$$y' = y - y^3$$



$$f(i) = \frac{1}{L} (V_{dc} - Ri)$$

$$V_{dc} = 2, R = 1, L = 1$$



Stability of Equilibrium Points - Local vs Global Stability

Locally stable $\rightarrow$ if trajectories that start around y^* converge to y^*

globally stable $\rightarrow$ if $y(t) \rightarrow y^*$ for all y_0 .

unstable $\rightarrow$ if trajectories that start very close to y^* move away from y^*

Condition for Local Stability and Assessment of stability for Non-linear systems

Let y^* be the Equilibrium

Goal:- Analyse solutions that start around y^*
by linearizing $\frac{dy}{dt} = f(y)$ about y^*

Let Δy denote a small deviation from y^* $\Rightarrow \Delta y(0)$ is the deviation of init cond from y^*
 y^* is locally stable $\rightarrow$ if $\Delta y^{(t)} \rightarrow 0$ as $t \rightarrow \infty$.

using Taylor's series $f(y^* + \Delta y) \approx \underbrace{f(y^*)}_{=0} + \Delta y f'(y^*)$
 $\Rightarrow f(y^* + \Delta y) \approx \Delta y f'(y^*)$

$$\frac{d}{dt}(y^* + \Delta y) = \frac{d}{dt}(\Delta y).$$

Linearized ODE around y^* is $\frac{d\Delta y}{dt} = \underbrace{\Delta y f'(y^*)}_{\lambda}$ (sol: $e^{\lambda t}$)

if $\lambda > 0$, $\Delta y \rightarrow \infty$ as $t \rightarrow \infty \Rightarrow y^*$ is unstable if $f'(y^*) > 0$

if $\lambda < 0$ $\Delta y \rightarrow 0$ as $t \rightarrow \infty \Rightarrow y^*$ is stable if $f'(y^*) < 0$.

Idea of Linearization

